

CMPG325 – Computer Networks Individual Project

Milestone 1: Network Design – Tswalu Steel Fabrication

Field	Details
Student	MAAKE, B
Student Number	41432363
Project ID	CMPG325-2026-029
Client ID	CLI-029
Assigned Organisation	Tswalu Steel Fabrication (Rustenburg)
Industry	Manufacturing

1. Client Requirements

1.1 Project Purpose

The development of a computer network for Tswalu Steel Fabrication, Rustenburg is the focus of this project. The goal is to create the network to enable reliable communication between the necessary devices and meet the client's security and connectivity requirements. Another important aspect of the design is the adoption of IPv6 and IPv4, and the use of dual stack approach to run the network. The solution needs to also prevent unauthorised admin changes to network devices and to allow the visitors to get on the network without intruding on the company's internal resources. The proposed design will be designed and simulated using Cisco Packet Tracer.

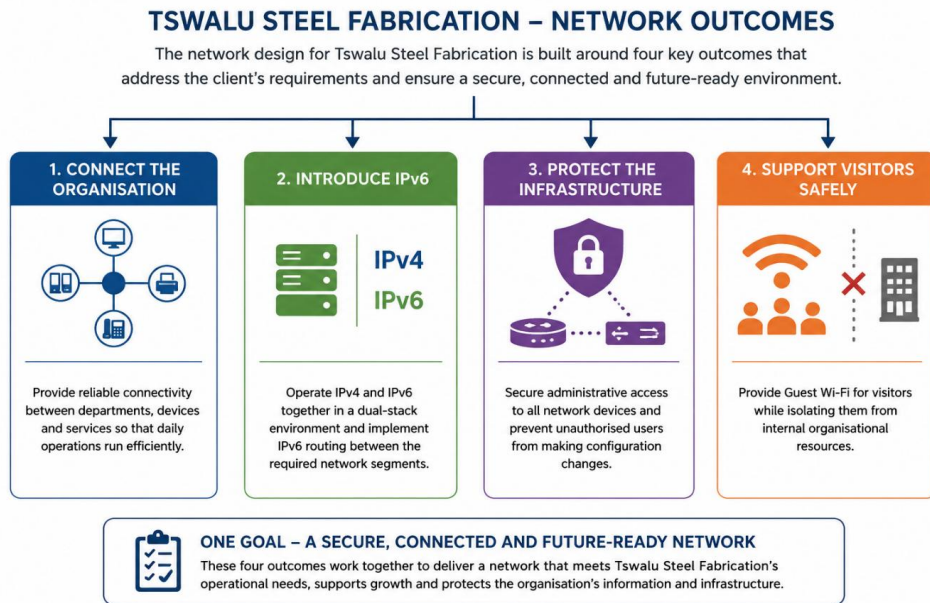


Figure 0: Network outcomes overview

1.2 Client Profile

Table 1 summarises the client and the key project parameters.

Item	Details
Organisation	Tswalu Steel Fabrication
Location	Rustenburg
Industry	Manufacturing
IPv4 Network	172.30.12.0/23
Technical Challenge	IPv6 dual-stack addressing & routing
Security Requirement	Protected device administration
Change Request	Guest Wi-Fi with internal-resource isolation

1.3 Network Requirements

The proposed network must provide a suitable foundation for the client's operations and must satisfy the requirements supplied in the project brief. The main requirements are:

- Provide connectivity between the appropriate network devices and nodes.
- Use 172.30.12.0/23 as the assigned IPv4 addressing space.
- Support IPv4 and IPv6 simultaneously.
- Provide IPv6 routing between the required network segments.
- Secure administration of network devices.
- Provide a separate network for visitors.
- Prevent guest users from accessing internal organisational resources.
- Produce a working implementation that can be tested in Cisco Packet Tracer.

1.4 Security Requirement

A security requirement is a requirement the client place on the security of the network devices in relation to changing them without authorization. The requirement will impact the management of routers and switches. This will be achieved through administrative access and use of proper controls to prevent unauthorised users from making configuration changes. The security configuration will be showcased during implementation and not considered as an additional feature to be implemented after the design of the network.

1.5 IPv6 Requirement

IPv6 dual-stack addressing and routing is the project's technical challenge.

The network will thus be able to run both IPv4 and IPv6. The necessary network segments will be addressed using suitable IPv6 addressing, and the assigned IPv4 network will be used for the IPv4 side of the design.

IPv6 routing will then be set up between the appropriate segments and verified to ensure that IPv6 communications are working properly. The project brief stipulates that this challenge must be set up, tested and shown to be working.

1.6 Guest Wi-Fi Requirement

The client's change request is to have Guest Wi-Fi for visitors but not allow them to access internal resources. The network will thus separate network traffic from visitors from other network traffic. The guest design will provide:

- Visitor device → Guest network → permitted access

while preventing:

- Visitor device → Internal organisational network

The exact method used to achieve this separation is established during the logical network design (see Section 4).

1.7 Addressing Approach

The assigned IPv4 block is 172.30.12.0/23. This address space is segmented based on the topology design network segments. The addressing plan was created after understanding the physical and logical network and the IP allocation is directly related to the devices and network segments needed. IPv6 addressing was designed in parallel with the IPv4 addressing plan and will be part of the final network that is a consistent dual-stack environment (see Section 5).

1.8 Design Boundary

The solution concentrates on the requirements given in the project brief:

- Tswalu Steel Fabrication's network connectivity.
- IPv4 addressing using the assigned address block.

- IPv6 dual-stack addressing and routing.
- Secure network-device administration.
- Guest Wi-Fi and separation from internal resources.
- Cisco Packet Tracer implementation and testing.

Features that do not contribute to these requirements are not added simply to make the topology larger or more complicated.

1.9 Requirement Summary

The design reduces to four main outcomes, summarised in Table 2.

Outcome	Description
Connect the organisation	Provide the required network communication between departments and devices.
Introduce IPv6	Operate IPv4 and IPv6 together and demonstrate IPv6 routing between segments.
Protect the infrastructure	Secure administrative access to network devices against unauthorised changes.
Support visitors safely	Provide guest Wi-Fi without exposing internal resources.

2. Proposed Design

The proposed solution addresses the client's requirements through network segmentation, dual-stack addressing, and a Layer 3 router providing inter-VLAN routing and security enforcement.

The design is built around three decisions:

- VLAN segmentation – the network is divided into five VLANs (Management & Administration, Production, Finance & HR, IT & Services, and Guest Wi-Fi), separating traffic by department and function.
- Router-on-a-stick routing – a single Layer 3 router (R1) connects to the core switch (SW1) over a trunk link, with one sub-interface per VLAN providing IPv4 and IPv6 default gateways and inter-VLAN routing.
- Guest isolation – the Guest Wi-Fi VLAN is kept logically and administratively separate from the four internal VLANs, satisfying change request CR3.

Table 3 traces each client requirement from Section 1 to the design decision that satisfies it.

Requirement	Design Response
Connectivity between departments	Five-VLAN segmentation with inter-VLAN routing via R1
172.30.12.0/23 addressing	VLSM subnetting of the assigned block across all VLANs (Section 5)
IPv4 + IPv6 dual-stack	Dual-stack sub-interfaces on R1 for every VLAN
Secured device administration	Restricted administrative access on R1 and SW1
Guest Wi-Fi, isolated from internal resources	Dedicated Guest VLAN 50 with no routing to internal VLANs

The following sections present the physical topology, logical topology, and IP addressing plan that implement this design.

3. Physical Topology

The physical topology below shows the devices and cabling implemented in Cisco Packet Tracer: a Layer 3 router (R1), a Layer 2 switch (SW1), PCs for Administration, Production, Finance, IT and Office use, a server, and a wireless access point serving the isolated Guest VLAN.

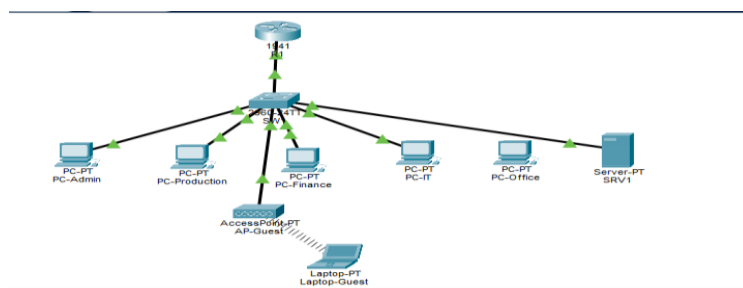


Figure 1: Physical topology (Cisco Packet Tracer)

4. Logical Topology

The logical topology shows how the Tswalu Steel Fabrication network is divided into separate network segments. The design is based on separation of different types of users and services by means of VLANs and the provision of communication between the necessary segments through routing. A separate guest network is provided to help prevent visitors' devices from accessing internal resources. The network supports both IPv4 and IPv6 as part of the required dual-stack implementation.

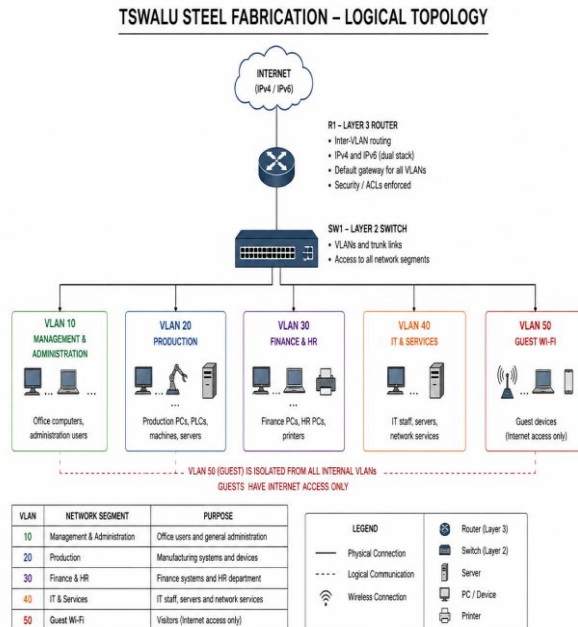


Figure 2: Proposed logical topology

5. Addressing Plan

The tables below present the IPv4 and IPv6 addressing plan for all VLANs in the Tswalu Steel Fabrication network.

4.1 IPv4 Addressing

VLAN	Segment	Network / Prefix	Usable Range	Gateway
10	Management & Administration	172.30.12.64/27	172.30.12.65 – .94	172.30.12.65
20	Production	172.30.12.0/26	172.30.12.1 – .62	172.30.12.1
30	Finance & HR	172.30.12.128/28	172.30.12.129 – .142	172.30.12.129
40	IT & Services	172.30.12.144/28	172.30.12.145 – .158	172.30.12.145
50	Guest Wi-Fi	172.30.12.96/27	172.30.12.97 – .126	172.30.12.97

4.2 IPv6 Addressing (Dual-Stack)

VLAN	Segment	IPv6 Prefix	IPv6 Gateway
10	Management & Administration	2001:db8:acad:10::/64	2001:db8:acad:10::1
20	Production	2001:db8:acad:20::/64	2001:db8:acad:20::1
30	Finance & HR	2001:db8:acad:30::/64	2001:db8:acad:30::1
40	IT & Services	2001:db8:acad:40::/64	2001:db8:acad:40::1
50	Guest Wi-Fi	2001:db8:acad:50::/64	2001:db8:acad:50::1

Addressing Summary

- IPv4 address block (assigned): 172.30.12.0/23
- IPv4 allocation method: VLSM (variable-length subnet masking)
- IPv6 design prefix (documentation prefix): 2001:db8:acad::/48

- IPv6 allocation: one /64 per VLAN
- Routing device: R1 (Layer 3 router, router-on-a-stick via sub-interfaces)
- Network model: IPv4 + IPv6 dual-stack

Notes

- The IPv4 block 172.30.12.0/23 is provided in the project brief.
- The VLAN structure and IPv6 prefix 2001:db8:acad::/48 are proposed design decisions for this project.
- VLAN 50 (Guest Wi-Fi) is isolated from all internal VLANs, satisfying change request CR3.
- All VLANs support IPv4 and IPv6 (dual-stack).

6. GitHub Portfolio of Evidence

Project evidence is maintained in the GitHub repository below and organised using the folder structure shown in Table 4.

Folder / File	Contents
configurations/	Router_R1_Config.txt and Switch_SW1_Config.txt
diagrams/	Network topology diagrams
docs/	Design_Document.pdf, Network_requirements.pdf and Reflection.md
ip-addressing/	IP addressing plan (IPv4/IPv6)
packet-tracker/	Tswalu_Network.pkt Cisco Packet Tracer project
screenshots/	Configuration, verification, connectivity and guest-isolation evidence
README.md	Project overview, network design, addressing, testing and evidence guide
troubleshooting/	Troubleshooting notes, diagnostic evidence, and problem-resolution screenshots

Repository: [blessedmaake18-bee/Tswalu_Network: Network design for Tswalu Steel Fabrication \(IPv6 dual-stack\)](#)

GitHub Repository Structure

Tswalu_Network/

```
|— configurations/
|   |— Router_R1_Config.txt
|   |— Switch_SW1_Config.txt
|— diagrams/
|— docs/
|   |— Design_Document.pdf
|   |— Network_requirements.pdf
|   |— Reflection.md
|— ip-addressing/
|— packet-tracker/
|   |— Tswalu_Network.pkt
|— screenshots/
|— troubleshooting/
|— README.md
```